Assignment 1: Questions, provenance and one variable

Statistical Thinking for Data Science • First Semester, 2026/2027

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Covering Weeks 1 to 4 • data science, answerable questions, where data came from, describing one variable

How to hand this in

Work on your own. Show your working: an answer with no working earns no marks, and a correct number explained badly earns less than a wrong number reasoned carefully. Write your **name, index number and programme** at the top of the first page.
Submit a single PDF through the link announced in class, and bring the paper copy to the next session. **The deadline will be communicated in class and posted on the course page.** Marked out of 50.
If you used any AI tool, add one line at the end saying which and what for. That line costs you nothing. An undeclared tool is academic dishonesty.

Part A: provenance and measurement (20 marks)

1. A contractor is paid to monitor a river near a mine. Each working morning at 08:00 a technician takes one water sample from the bridge, and the laboratory reports pH, turbidity and three metals. The contract has run for two years.
   (a) Name the unit of observation, the population of interest, and the sample actually obtained. (3)
   (b) Classify each of the five reported variables by level of measurement. (2)
   (c) List three parts of the population that this design can never observe, and for each say whether you expect it to make the river look better or worse than it is. (3)
   (d) The contractor's report gives a single mean pH for the two years. State two questions you would ask before quoting that number in a regulatory submission. (2)
2. Turn each vague question into one that data could answer, and say what you would have to measure. Keep each answer to two sentences. (6)
   (a) Is mining damaging the environment in Tarkwa?
   (b) Are our students struggling this semester?
   (c) Is the new conveyor better?
3. A dataset arrives with a column named `status` containing the values 1, 2, 3 and 99. Describe exactly what you would do before computing anything with it, and explain what could go wrong if you did not. (4)

Part B: describing one variable (20 marks)

Twenty daily pH readings from the monitoring point above:

6.1, 6.3, 6.4, 6.5, 6.5, 6.6, 6.7, 6.8, 6.8, 6.8, 6.9, 6.9, 7.0, 7.1, 7.2, 7.4, 7.6, 7.9, 8.3, 9.1

4. Compute the mean, median, mode, range, quartiles, interquartile range and standard deviation. (8)
5. Apply the $1.5 \times$ IQR rule and state which readings, if any, are flagged. (3)
6. The mean exceeds the median. Say what that tells you about the shape, and name the readings responsible. (3)
7. Recompute the mean, median and standard deviation with the largest reading removed. Which summary moved most, and what general lesson does that carry? (4)
8. The regulator wants one number for the month. Choose it, and write the single sentence you would submit alongside it. (2)

Part C: in your own field (10 marks)

In **half a page, handwritten**, describe one dataset that exists in your own programme of study: what it measures, who collected it and why, and one group it systematically misses. Then state one question it can answer and one it cannot.

Marks are for the specificity. "Data about mining" earns two; "the daily blast vibration logs kept by the geotechnical section at a named pit, which only record blasts the monitor was switched on for" earns ten.

Total: 50 marks. Parts A and B may be typed; Part C must be in your own handwriting.